

Nikhil Talagana

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Summary

PostgreSQL & Oracle Database Administrator with 2+ years of experience in managing on-premises and AWS environments. Skilled in backup & recovery, performance tuning, high availability, and disaster recovery. Hands-on experience with AWS RDS, EC2, and database automation using Python. Additionally, experienced in leveraging AI/GenAI technologies (RAG, LangChain) to build intelligent automation and monitoring solutions.

Skills

Databases	:	PostgreSQL, Oracle
Operating Systems	:	Linux
Cloud (AWS)	:	RDS, Aurora, EC2, S3, IAM, CloudWatch, Lambda, SSM, CloudTrail, Bedrock, Terraform
Core DBA Skills	:	High Availability (HA), Disaster Recovery (DR), Performance Tuning, Replication, Backup & Recovery
Scripting Languages	:	Python, Shell
Tools	:	DBeaver, PL/SQL Developer, PgAdmin
Gen AI	:	RAG, LLM, LangChain, LangGraph

Professional Experience:

PostgreSQL Database Administrator
Vitech Systems Asia Pvt Ltd.

March 2024 – March 2026
Hyderabad, Telangana

PostgreSQL DBA (On-Premises & AWS Cloud)

- Installed, configured, and maintained PostgreSQL on Linux and Windows environments, including cluster initialization and configuration management.
- Created and managed databases, schemas, users, and roles with proper access control using `pg_hba.conf` and privilege management.
- Tuned PostgreSQL performance through configuration optimization, indexing strategies, query tuning, `VACUUM`, and `ANALYZE`.
- Performed backup and recovery operations including full, schema-level, and table-level backups using `pg_dump/pg_restore`, and implemented Point-in-Time Recovery (PITR) using WAL archiving.
- Managed high availability and streaming replication in PostgreSQL by creating standby servers using `pg_basebackup`, configuring replication parameters, monitoring replication lag, and performing manual failover and recovery operations.
- Executed PostgreSQL version upgrades using `pg_upgrade` with proper validation and rollback planning.
- Used tools like `pgBadger` for performance analysis and `pg_repack` for table/index maintenance with minimal downtime.
- Performed data masking/scrambling to secure sensitive data in non-production environments.

AWS (RDS/EC2/S3)

- Created and managed Amazon RDS and Aurora PostgreSQL instances, including provisioning, monitoring, patching, scaling, and version upgrades.
- Performed RDS snapshot creation, restoration, and cross-account snapshot sharing.
- Implemented Point-in-Time Recovery (PITR) and disaster recovery strategies in AWS environments.
- Launched and managed EC2 instances, including PostgreSQL client setup and connectivity to RDS/Aurora databases.
- Mounted and integrated Amazon S3 with EC2 (using tools like `s3fs`) for backup storage and data transfer.
- Managed database credentials securely using AWS Secrets Manager.

- Automated operational tasks using AWS Systems Manager (Run Command) and AWS Lambda for backup, restore, and maintenance workflows.
- Monitored database and infrastructure logs using Amazon CloudWatch and configured alerts for performance and availability.
- Tracked and audited AWS activities using AWS CloudTrail for security and compliance.
- Generated performance reports for RDS PostgreSQL using pgBadger for monitoring and tuning insights.

Oracle Database Administration

- Installed and configured Oracle Database on Linux and Windows platforms, including database creation using DBCA (GUI) and silent mode with response files.
- Configured and managed Oracle networking components, including Listener and TNS (using Net Manager/NETCA).
- Managed Oracle memory architecture, including understanding and monitoring of SGA and PGA components.
- Created and managed database users, roles, and privileges at schema and object levels, including read-only and application-specific access.
- Configured and maintained Undo Tablespaces and performed routine database maintenance activities, including defragmentation.
- Managed ASM disk groups and storage for database environments.
- Performed backup and recovery operations using RMAN, along with logical backups using Data Pump (EXPDP/IMPDP).
- Executed database upgrades using DBUA with proper validation and post-upgrade checks.
- Performed data masking/scrambling to secure sensitive data in non-production environments.
- Monitored database performance and resolved alerts using Oracle Enterprise Manager (OEM), including tablespace utilization, availability issues, and high CPU usage.
- Worked with Oracle Multitenant architecture (CDB/PDB) for database management.
- Gained exposure to high availability and disaster recovery solutions using Oracle Data Guard.
- Exposure to Oracle RAC environments, including monitoring and basic administrative support.

Automation & AI-Driven Enhancements

- Developed Python-based monitoring scripts to capture PostgreSQL lock details at regular intervals for proactive troubleshooting and issue resolution.
- Implemented cross-account RDS snapshot copy for disaster recovery (DR) and environment refresh activities.
- Enhanced pg_restore automation by dynamically calculating parallel jobs based on RDS instance class and database size, improving restore performance.
- Automated Blue-Green deployment process, including Green RDS provisioning and controlled switchover, enabling low-downtime PostgreSQL releases.
- Developed an AI-powered Jira Bot to suggest resolutions for newly created tickets, improving issue handling efficiency.
- Automated ingestion of closed Jira tickets into a vector database to continuously improve AI-based recommendations and analysis.
- Generated weekly AI-driven metrics comparing developer fixes vs AI-suggested solutions to measure effectiveness.
- Built an AI-assisted validation framework to analyze Blue-Green deployment packages and ensure database compatibility before deployment.

Education

Bachelor of technology

University College of Engineering Narasaraopet, JNTUK
CGPA: 6.7

2019-2023
AP, India

Professional Training:

- AWS Cloud – Gana Technologies, Hyderabad
(Hands-on training on EC2, RDS, S3, IAM, CloudWatch, VPC basics)